

PIML-UrbanHeat: A Physics-Informed Neural Framework for Counterfactual Urban Heat Mitigation in Humid-Tropical Coastal Cities

First Interim Project Evaluation

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Presentation Outline

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Background

- Rapid urbanization in Kerala's humid-tropical coastal cities is increasing the Urban Heat Island (UHI) effect.
- Built-up surface expansion and vegetation loss increase Land Surface Temperature (LST).
- High relative humidity further increases thermal discomfort in coastal urban areas.
- Existing machine learning models can predict temperature but may produce physically inconsistent results.

Project Idea

- PIML-UrbanHeat is a physics-informed AI platform for modelling urban land surface temperature.
- The system combines satellite remote sensing with Physics-Informed Neural Networks (PINNs).
- It supports counterfactual “what-if” simulations for urban heat mitigation strategies.

Technology Direction

- Physics-Informed Neural Networks (PINNs)
- Satellite Remote Sensing
- Surface Energy Balance modelling
- Google Earth Engine and ERA5-Land data

Relevance and Motivation

- Urban heat causes thermal stress and affects public health.
- High humidity in Kerala's coastal cities increases discomfort during extreme temperature conditions.
- Municipal authorities invest in tree planting, parks, green roofs and cool-roof strategies.
- There is limited capability to estimate how much cooling each intervention will provide before implementation.
- Satellite data and AI can support better urban climate planning and decision-making.
- Physics-informed learning can improve the physical consistency of temperature predictions.

Motivation Statement

- Develop an intelligent and physically consistent system for predicting and mitigating urban heat.
- Enable planners to evaluate “what-if” mitigation strategies before implementation.

Existing / Conventional Approach

- Random Forest models
- XGBoost models
- CNN-based temperature prediction
- Physics-based simulators
- Historical satellite observation studies

Limitations / Gaps

- ML models can behave as black boxes
- Possible violation of physical laws
- Physics simulators are computationally expensive
- Existing Kerala studies are mainly retrospective
- Limited support for real-time intervention simulation

Example Scenario 1

Situation

An urban planner wants to identify areas with high Land Surface Temperature and understand the influence of vegetation and built-up surfaces on urban heat.

Sample Input

- Landsat 8/9 thermal imagery
- Sentinel-2 multispectral imagery
- NDVI, NDBI and Albedo values
- Weather and climate variables

Example Scenario 1: Expected Output

Expected Output

- Predicted Land Surface Temperature map
- Urban heat distribution across selected areas
- Identification of potential urban heat hotspots
- Physically consistent temperature predictions

Example Scenario 2

Situation

An urban planner wants to test whether increasing vegetation or applying cool roofs can reduce temperature in a selected ward.

Sample Input

- Selected city ward or urban area
- Existing surface parameters
- Modified NDVI values for green infrastructure
- Modified Albedo values for cool roofs

Example Scenario 2: Expected Output

Expected Output

- Predicted temperature change
- Cooling heatmap showing ΔT
- Comparison of different mitigation strategies
- Decision support for urban planning

Literature Survey

Paper / Work	Method Used	Key Contribution / Relevance
Raissi et al. (2019)	Physics-Informed Neural Networks	Introduced PINNs for solving forward and inverse problems using physical constraints.
Karniadakis et al. (2021)	Physics-Informed Machine Learning	Provides a foundation for combining physical laws with machine learning.
Urban LST using PINNs (2026)	PINN-based temperature modelling	Applies physics-informed learning to urban land surface temperature prediction.
Smart Urban Design with PINNs (2025)	PINNs and satellite data	Studies temperature reduction using green infrastructure and thermal data.
Kerala UHI Studies	Satellite remote sensing	Analyses land surface temperature and urban heat patterns in Kerala cities.
SEBAL	Surface Energy Balance	Provides parameterizations for radiation and heat flux estimation.

Research Gap

- Existing ML models may predict LST without enforcing thermodynamic conservation laws.
- Physics-based simulators require significant computational time.
- Existing Kerala UHI studies are mainly observational and retrospective.
- Limited systems support interactive counterfactual urban heat mitigation planning.
- Humid-tropical coastal conditions are not sufficiently addressed.

Proposed Novelty

- Humid-tropical coastal microclimate calibration
- Physics-constrained prediction using Surface Energy Balance loss
- SEBAL-style parameterization of heat fluxes
- Real-time “what-if” mitigation simulation
- Interactive urban heat planning for Kerala cities

Problem Statement

Background

Rapid urbanization in Kerala's humid-tropical coastal cities has increased the Urban Heat Island effect. Vegetation loss and built-up surface expansion increase land surface temperature, while high relative humidity intensifies thermal discomfort.

Existing machine learning models may produce physically inconsistent predictions, while physics-based numerical simulators require high computational time. Existing UHI studies in Kerala mainly analyse historical temperature patterns and do not provide counterfactual simulation capabilities.

Proposed Problem

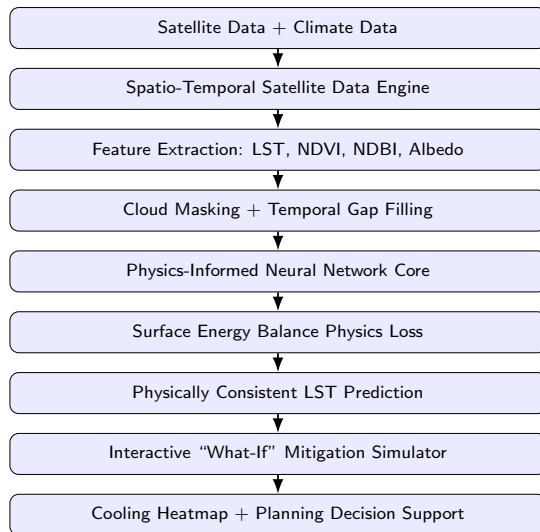
To develop a physics-constrained and computationally efficient platform that combines satellite remote sensing and Physics-Informed Neural Networks to predict urban heat and simulate mitigation strategies for humid-tropical coastal cities.

Project Objectives

- 1 Study existing urban heat prediction and mitigation approaches.
- 2 Collect and process satellite-derived thermal and spectral data.
- 3 Extract NDVI, NDBI and Albedo information.
- 4 Integrate climate variables from ERA5-Land data.
- 5 Develop a Physics-Informed Neural Network for LST prediction.
- 6 Incorporate Surface Energy Balance constraints.
- 7 Ensure physically consistent temperature predictions.
- 8 Develop counterfactual urban heat mitigation simulations.
- 9 Compare mitigation strategies such as green roofs and cool roofs.
- 10 Develop an interactive decision support interface for planners.

- ➊ **Satellite Data Collection:** Collect Landsat and Sentinel-2 satellite imagery.
- ➋ **Feature Extraction:** Calculate LST, NDVI, NDBI and surface Albedo.
- ➌ **Climate Data Integration:** Integrate air temperature, humidity and wind speed from ERA5-Land.
- ➍ **Data Preparation:** Apply preprocessing, cloud masking and temporal gap filling.
- ➎ **PINN Development:** Train a neural network to predict Land Surface Temperature.
- ➏ **Physics Constraint:** Incorporate Surface Energy Balance residual into the loss function.
- ➐ **Counterfactual Simulation:** Modify surface parameters for urban mitigation scenarios.
- ➑ **Interactive Visualization:** Display predicted cooling effects using maps and heatmaps.

Proposed System Architecture



Dataset Requirements

Dataset / Data Type	Required Fields / Information	Possible Source
Landsat Data	Thermal Band, Land Surface Temperature and QA information	Landsat 8/9
Sentinel-2 Data	Multispectral bands for NDVI, NDBI and Albedo	Sentinel-2
Climate Data	Air temperature, humidity and wind speed	ERA5-Land
MODIS LST	Thermal composites for temporal gap filling	MODIS MOD11A2
Geospatial Data	City and ward boundaries	Public GIS datasets

Study Area

Selected humid-tropical coastal urban areas in Kerala will be considered for model development and evaluation.

- ① **Spatio-Temporal Satellite Data Engine** – collects and processes satellite imagery.
- ② **Feature Extraction Module** – calculates LST, NDVI, NDBI and Albedo.
- ③ **Cloud and Gap Filling Module** – handles cloud cover and temporal data gaps.
- ④ **Climate Data Module** – integrates ERA5-Land climate variables.
- ⑤ **PINN Core** – predicts Land Surface Temperature using neural networks and physics constraints.
- ⑥ **Surface Energy Balance Module** – calculates energy balance components.
- ⑦ **Counterfactual Simulation Module** – evaluates green roofs, cool roofs and urban greening.
- ⑧ **Interactive Planning Interface** – visualizes cooling predictions and heatmaps.

Software Requirements

Development Tools

- Python
- VS Code / Jupyter Notebook
- Git and GitHub
- Google Earth Engine Python API
- Next.js

Libraries / Frameworks

- PyTorch
- DeepXDE
- Google Earth Engine API
- Geospatial data processing libraries
- Mapbox GL JS

Core Technology

Physics-Informed Neural Networks, satellite remote sensing and Surface Energy Balance modelling form the main technical direction.

Hardware Requirements

Minimum Requirements

- Laptop or desktop computer with a suitable processor.
- Minimum 8 GB RAM.
- Sufficient storage for satellite and geospatial datasets.
- Stable Internet connection for cloud-based data access.

Recommended / Optional Requirements

- 16 GB RAM or above.
- GPU support for faster neural network training.
- Cloud computing resources for large-scale experimentation.
- No specialized hardware is required for basic prototype development.

Work Completed So Far

- Project idea and research scope have been identified.
- The proposed system has been defined as PIML-UrbanHeat.
- Major satellite, climate and surface parameters have been identified.
- Physics-Informed Neural Networks have been selected as the core modelling approach.
- Surface Energy Balance constraints have been identified for physically consistent prediction.
- Literature related to PINNs, urban heat, satellite remote sensing and SEBAL has been reviewed.
- The counterfactual “what-if” simulation concept has been defined.

Current Stage

Project proposal, literature study and system design are being developed. Implementation results are not included at this stage.

Current Status

- Project scope defined
- Study area identified
- Core architecture planned
- Literature survey completed
- Major modules identified

Initial Results

- No quantitative model results are available yet.
- Dataset collection and preprocessing are planned as the next stages.
- PINN implementation and baseline comparison will be performed during development.

Evaluation Plan

Component	Metric	Purpose
LST Prediction	R^2 , RMSE, MAE	Measures temperature prediction accuracy
Physics Consistency	SEB residual magnitude	Measures consistency with Surface Energy Balance
Counterfactual Simulation	Predicted ΔT ranges	Evaluates cooling predictions under interventions
Baseline Comparison	RF, XGBoost, CNN vs PINN	Measures the benefit of physics-informed learning
System Performance	Response and processing time	Measures practical efficiency

Project Completion Plan

Period	Planned Work
Weeks 1–2	Literature review, requirement analysis and scope finalisation
Weeks 3–4	Satellite dataset identification and data collection
Weeks 5–7	Data preprocessing, cloud masking and feature extraction
Weeks 8–10	LST prediction baseline using RF, XGBoost and CNN
Weeks 11–13	Development of Physics-Informed Neural Network
Weeks 14–16	Surface Energy Balance physics loss integration
Weeks 17–20	Counterfactual mitigation simulation development
Weeks 21–24	Interactive visualization, testing and evaluation
Weeks 25–28	Final integration, documentation, demonstration and project report

Expected Outcome

- A working prototype of the PIML-UrbanHeat platform.
- Physically consistent Land Surface Temperature prediction.
- Integration of satellite remote sensing and climate data.
- Physics-Informed Neural Network with Surface Energy Balance constraints.
- Counterfactual simulation of urban heat mitigation strategies.
- Real-time prediction of cooling effects.
- Interactive ΔT cooling heatmaps.
- Decision support for urban planners and municipal authorities.
- Comparison with conventional machine learning models.

- Satellite imagery may contain cloud cover and missing data.
- Landsat revisit frequency can create temporal gaps.
- Climate and satellite data may have different spatial resolutions.
- PINN training can require significant computational resources.
- Counterfactual intervention scenarios cannot be directly observed before implementation.
- Model predictions may contain uncertainty.
- Results should support urban planners rather than replace expert judgement.
- Urban planning decisions should consider environmental and social impacts.

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Summary

PIML-UrbanHeat proposes a physics-informed AI system for understanding and mitigating urban heat in humid-tropical coastal cities. The system combines satellite remote sensing, climate data and Physics-Informed Neural Networks to generate physically consistent Land Surface Temperature predictions.

The proposed “what-if” simulator enables urban planners to evaluate strategies such as green roofs, cool roofs and urban greening before implementation.

Immediate Next Steps

- Finalise study areas and datasets.
- Collect and preprocess satellite data.
- Implement baseline machine learning models.
- Develop the PINN with physics constraints.
- Develop and evaluate the mitigation simulator.

Thank You